

PROJECT PROGRESS REPORT 1

Data Integration and Exploratory Analysis

For Port Authority Bus Terminal

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Contributions by the Team

Group Member	Contribution
Hassan Siddiqui	Contributed in sampling techniques and presentation
Iftekhar Kaiser Shimul	Drafted the report and explored external datasets for the project
Pritishma Shakya	Drafted the report and explored tools & technologies
Raghav Rishi	Contributed in data integration phase
Yusuf Oli Rahat	Contributed to data cleaning and exploration

1. AVAILABLE DATA SOURCES

We received four datasets from Port Authority covering traffic patterns from 2013 to 2025 which are summarized in the table below:

Dataset Name	Number of Records	What It Contains
Traffic Data	5,383,378 records	Hourly vehicle counts at bridges/tunnels
Bus Terminal	2,784 records	Weekly bus counts by carrier
Passenger Data	2,786 records	Weekly passenger counts by carrier
Speed Data	192 records	Monthly average speeds at facilities

What Each Dataset Tells Us

Traffic Data

The Traffic dataset is the primary dataset in this project, containing 5.4 million hourly records across 30 columns spanning September 2013 to May 2025. It tracks every vehicle — cars, trucks, and buses — crossing all 9 Port Authority facilities including bridges, tunnels, and Staten Island crossings, recording how each driver paid: cash, E-ZPass, or not at all. This dataset serves as the foundation for revenue tracking, congestion planning, and operational analysis.

Bus Terminal and Passenger Data

The PABT Bus and Passenger datasets contain weekly records tracking bus and passenger volumes at the Port Authority Bus Terminal, broken down by carrier including NJ Transit, Greyhound, and Coach USA. Covering December 2020 to May 2025 (4.5 years), the bus dataset holds 2,784 records while the passenger dataset contains 2,786 records. These datasets are particularly valuable for analyzing the recovery of public bus transit following the COVID-19 pandemic.

Speed Data

The Facility Mobility Speeds dataset contains 192 records measuring actual traffic speeds across the 6 Port Authority facilities in both directions, comparing them against free-flow speeds (the ideal speed under no-traffic conditions). The difference between the two (Delta) reveals which facilities experience the worst congestion. While highly informative for congestion analysis, this dataset is currently limited in scope, covering only a short time window which restricts long-term trend analysis.

2. EXPLORATORY DATA ANALYSIS

Python was used extensively in this project to clean and analyze the data. The table below displays comprehensive statistics calculated on the original datasets before aggregation. These show the characteristics of individual records (hourly traffic, weekly bus/passenger counts).

Traffic Dataset Statistics (5.4 Million Hourly Records)

Statistics on hourly traffic records across all lanes, facilities, and time periods can be seen in the table below:

Variable	Mean	Median	Mode	Variance	Std Dev	Min	Max	Range
TOTAL (vehicles/hour)	270.17	199	1	67,914.80	260.60	0	3,938	3,938
CLASS 1 (Autos)	245.79	175	1	61,680.24	248.36	0	3,148	3,148
CLASS 2 (Small trucks)	6.82	2	0	209.08	14.46	0	256	256
CLASS 3	2.18	0	0	26.92	5.19	0	237	237
CLASS 5 (Buses)	6.88	0	0	411.14	20.28	0	471	471
CASH (transactions)	36.36	3	0	2,807.30	52.98	0	817	817
EZPASS (transactions)	218.75	138	1	53,679.51	231.69	0	3,936	3,936
VIOLATION	15.07	7	0	512.78	22.64	0	789	789
HOUR (0-23)	11.69	12	9	46.07	6.79	0	23	23
DAY (1-7)	3.99	4	1	4.00	2.00	1	7	6
LANE	17.79	10	4	386.98	19.67	1	74	73
TIME (HHMM)	1,168.66	1,200	900	460,662.40	678.72	0	2,300	2,300
FAC (Facility code)	5.14	6	7	7.61	2.76	1	9	8
YEAR	2,017.87	2,018	2,013	11.22	3.35	2,013	2,025	12
MONTH	6.40	6	1	11.97	3.46	1	12	11

Key Insights from Traffic Statistics

The average hourly lane volume is 270 vehicles with a peak of 3,938 vehicles recorded in a single hour, reflecting high variability across lanes and time periods. The median of 199 is lower than the mean of 270, indicating a right-skewed distribution where most hours have moderate traffic but occasional high-volume surges pull the average up. The mode of 1 confirms that many lanes experience very light traffic during off-peak hours, with most lanes recording zero vehicles for heavier vehicle classes.

Passenger cars dominate at 91% of all traffic, averaging 246 out of 270 vehicles per hour, meaning they essentially define total traffic volume. E-ZPass accounts for 86% of all toll transactions at around 219 per hour, compared to just 36 cash payments, reflecting strong adoption of electronic tolling. Toll violations average 15 per hour but are highly variable with a standard deviation of 23, suggesting significant differences in violation behavior across facilities and time periods. Overall, traffic is broadly spread across all hours and days with an average hour of around noon and average day falling on Wednesday.

PABT Bus Terminal and Passenger Statistics

The table below shows statistics on weekly bus and passenger volumes by carrier (2,784 weekly records from Dec 2020 - May 2025):

Variable	Mean	Median	Mode	Variance	Std Dev	Min	Max	Range
Bus Volume	221.71	29	0	377,395.59	614.33	0	2,577	2,577
Passenger Volume	6,177.60	796	0	325,197,779.30	18,033.24	0	90,225	90,225

Key Insights from PABT Statistics

Bus Volume:

Bus volume averages 222 buses per carrier per week, but the median of just 29 reveals a highly right-skewed distribution where a few large carriers dominate while many smaller carriers operate infrequently. Major carriers like NJ Transit handle thousands of buses per week, with a recorded maximum of 2,577 buses for a single carrier in one week, while many smaller carriers have weeks with zero buses. The standard deviation of 614 is 2.8 times the mean, indicating extreme variability and confirming that bus volume is heavily concentrated among a small number of high-volume carriers.

Passenger Volume:

Passenger volume averages 6,178 passengers per carrier per week, but the median of just 796 reveals an extreme right-skewed distribution that is even more pronounced than the bus volume data. The average passenger to bus ratio works out to approximately 28 passengers per bus, though this varies significantly by carrier and route type. One carrier transported a maximum of 90,225 passengers in a single week, while many carriers had weeks with zero passengers due to limited or no service, resulting in a standard deviation of 18,033 which is 2.9 times the mean. This extreme

concentration confirms that a small number of major carriers handle the vast majority of passenger traffic at the Port Authority Bus Terminal.

Speed Data Statistics

The table below shows statistics on facility speeds (192 records covering 6 facilities × 2 directions from Jan 2024 - May 2025):

Variable	Mean	Median	Mode	Variance	Std Dev	Min	Max	Range
Free-flow Speed (mph)	42.02	44.90	44.90	29.04	5.39	32.00	47.90	15.90
Actual Avg Speed (mph)	35.73	39.20	48.50	137.84	11.74	13.20	49.50	36.30
Delta (slowdown in mph)	-6.52	-4.90	2.00	79.28	8.90	-27.30	4.10	31.40
Facility Order	3.50	3.50	1.00	2.93	1.71	1.00	6.00	5.00

Key Insights from Speed Statistics

The average free-flow speed is 42 mph while the average actual speed is 36 mph, representing a 15% slowdown and an average congestion delta of 6.52 mph below free-flow. The most common free-flow speed is 44.90 mph, reflecting similar design speeds across tunnels and bridges, though actual speeds vary dramatically with a standard deviation of 11.74 mph depending on time and facility.

Congestion ranges from severe to non-existent across the dataset. The worst recorded slowdown is 27.30 mph below free-flow, likely reflecting severe rush hour conditions at the Lincoln or Holland Tunnel, while the best conditions show speeds 4.10 mph above free-flow as drivers exceed posted limits during light traffic. Recorded speeds span from a minimum of 13.20 mph, likely the Lincoln Tunnel during peak congestion, to a maximum of 49.50 mph, likely the Outerbridge Crossing during off-peak hours. Interestingly the delta mode is +2.00 mph, meaning many time periods see traffic moving slightly faster than the designed free-flow speed, suggesting that off-peak conditions are quite common across the dataset.

Comparison: Raw Data vs Aggregated Data

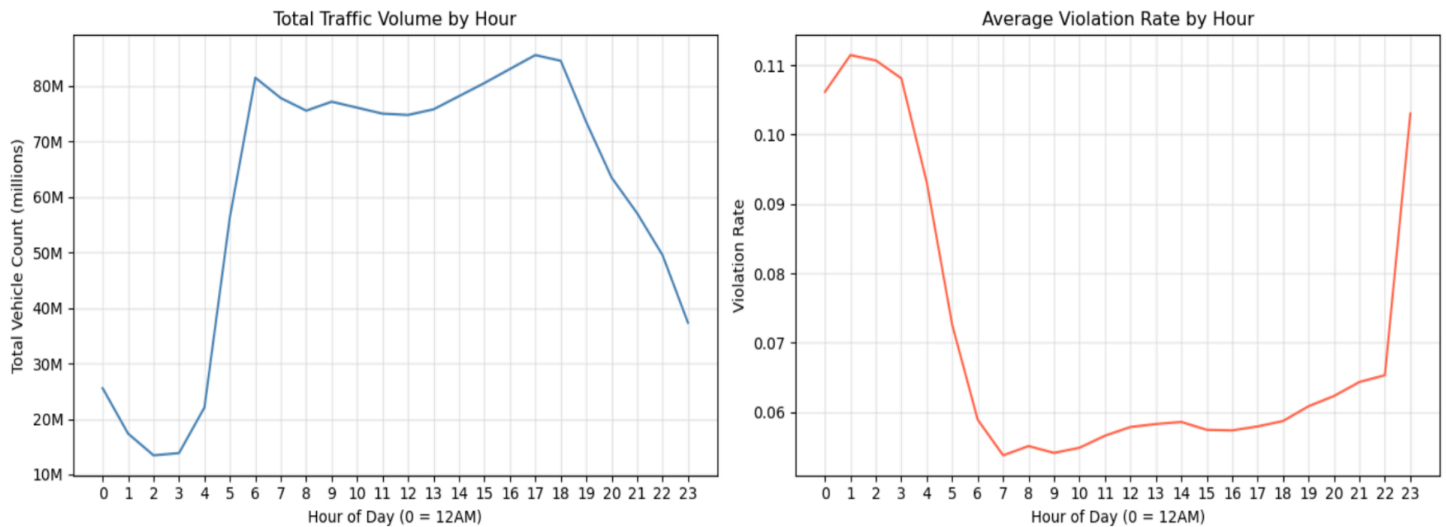
The statistics above are from the original datasets (hourly traffic, weekly PABT). After aggregation to the monthly level for the integrated dataset, the statistics change dramatically:

Metric	Original Granularity	Aggregated Monthly
Traffic Mean	270.17 vehicles/hour	9,897,811 vehicles/month
Traffic Median	199 vehicles/hour	10,084,757 vehicles/month
Bus Mean	221.71 buses/week	11,958 buses/month
Passenger Mean	6,177.60 passengers/week	378,893 passengers/month

Note: These statistics are from the original hourly/weekly datasets. The final integrated dataset aggregates these to monthly totals.

Traffic Patterns We Discovered

Hourly Traffic Analysis



Total Traffic Volume by Hour

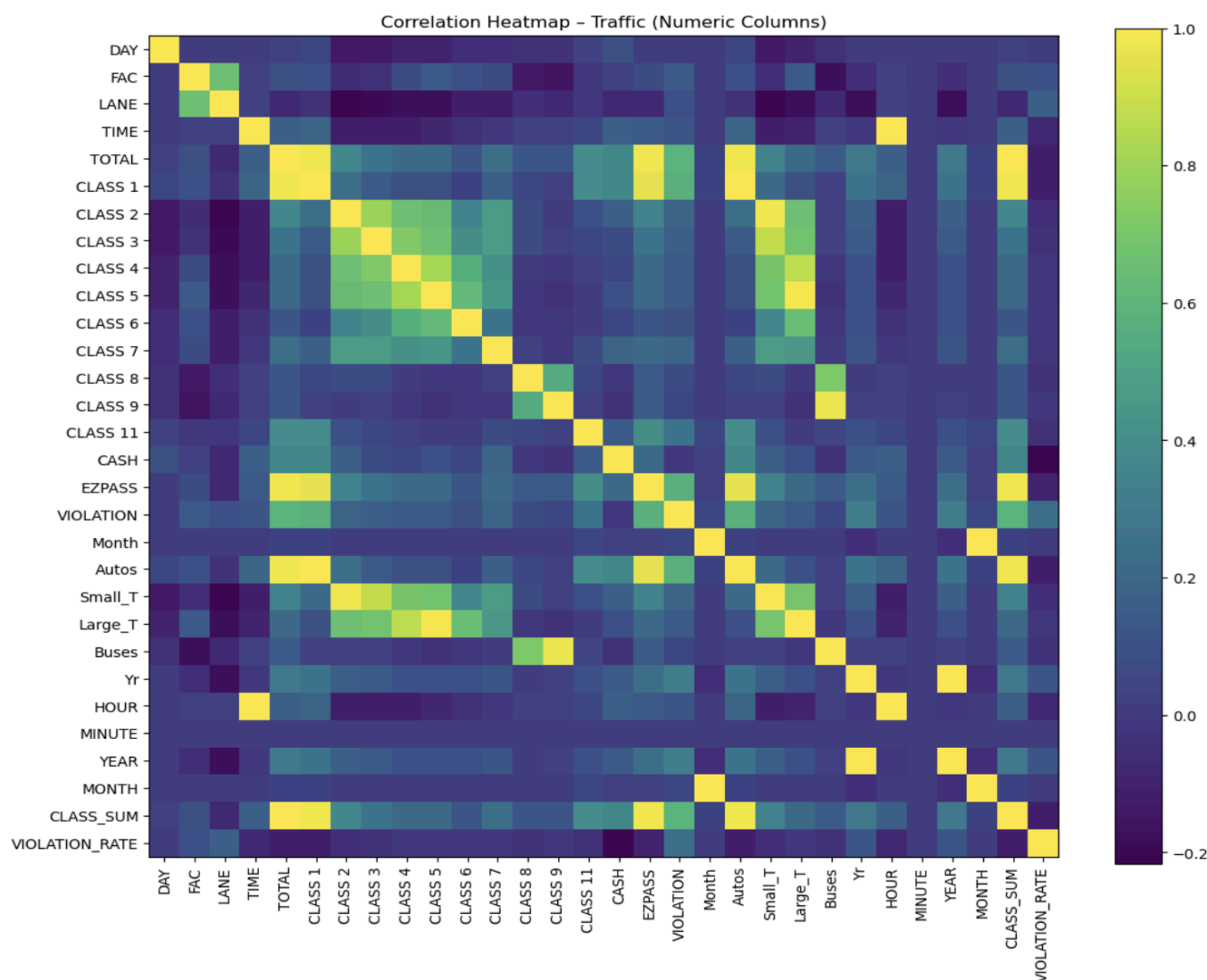
Traffic is at its lowest around 3AM with about 13 million vehicles, then rises sharply by 6AM reaching around 81 million, reflecting the morning rush hour. Volume stays fairly steady throughout the day between 75 to 85 million vehicles before dropping again after 7PM. From a business perspective, this tells us that toll infrastructure and staffing need to be fully operational during daytime hours, while the late night window between 1AM and 3AM presents the best opportunity for maintenance, system updates, or staff rotations with minimal disruption.

Average Violation Rate by Hour

Violation rate is highest between 1AM and 2AM at around 11%, when overall traffic is low, and drops to its lowest point of around 5% at 7AM when traffic is at its peak. It stays low and flat at around 6% throughout the daytime before rising sharply again after

10PM. This is a direct revenue and enforcement concern, as a higher proportion of vehicles are not paying correctly during nighttime hours despite fewer vehicles being on the road. Targeted enforcement or improved detection systems during these hours could recover significant lost toll revenue.

Relationships Between Variables



The correlation heatmap reveals that total traffic has a strong positive correlation of over 90% with passenger cars and E-ZPass usage, meaning they move together consistently. Cash payments show a moderate and weakening correlation with traffic as electronic tolling becomes more dominant. This tells us that investing further in E-ZPass infrastructure would have the highest return, and reducing cash toll lanes in favor of electronic ones would improve throughput and cut operational costs. Truck and bus

classes show weak correlations with each other and with general traffic, meaning they require separate traffic management strategies and pricing models rather than being treated the same as passenger cars.

Summary of Key Insights

Passenger cars account for 91% of all traffic, averaging 246 out of 270 vehicles per hour, which explains their strong correlation with total volume. Traffic volume varies dramatically throughout the day, with 6 times more traffic at 6 PM compared to 3 AM, and exhibits a clear rush hour pattern with a sharp morning increase starting at 5 AM and an evening peak between 6-7 PM. E-ZPass dominance is evident, with 85-90% of all toll payments made electronically, reflecting widespread adoption of the system. There is also a notable violation of paradox. The lowest traffic periods (late night/early morning) actually have the highest violation rates, suggesting different driver behavior during off-peak hours. Finally, the high variability in traffic, with a standard deviation (261) nearly equal to the mean (270), indicates that hourly volumes are highly unpredictable and can swing wildly depending on time, day, and conditions.

3. CLEANING THE DATA

We used Python in order to clean the data. A detailed description of the ways in which we addressed this is explained below:

Python Codes: [Data Exploration.pdf](#)

Problem 1: Missing Information

Vehicle Class	Missing Records	% Missing
CLASS 6	1,354,198	25.2%
CLASS 7	1,162,245	21.6%
CLASS 11	1,011,351	18.8%
CLASS 4	977,073	18.2%

The traffic data tracks 11 different vehicle types (cars, small trucks, buses, etc.). Many records were missing information about certain vehicle types as seen in the above table.

To resolve this issue, when a vehicle type was blank, we assumed it meant zero vehicles of that type. A total of 8,511,288 cells were fixed, with missing values being converted to zeros.

Problem 2: Null Values

The data uses different codes for missing values throughout the 5.4 million records like empty cells (just blank), the words "NULL", "NaN" (Not a Number) and spaces or whitespace characters.

To fix this, we told Python to treat all these different codes the same way. Everything was standardized to a consistent format during data import and we used the `na_values` parameter to catch all variations.

Problem 3: Messy Text

Facility names and other text had inconsistent formatting across thousands of records like extra spaces: " Lincoln Tunnel " vs "Lincoln Tunnel", mixed capitalization and multiple spaces between words. This made it hard to count and group the same facilities.

To fix this problem, we removed extra spaces from beginning and end (strip) and made all text uppercase for consistency. We also collapsed multiple spaces into single spaces. For example: " lincoln tunnel " was converted to "LINCOLN TUNNEL". These were applied to all text columns: facility names, lane modes, directions.

Problem 4: Extreme Values (Outliers)

Statistic	Value
Normal range (25th-75th percentile)	143 to 357 vehicles/hour
Outlier threshold	Above 761 vehicles/hour
Number of outliers found	339,366 records (6.3%)
Highest outlier	3,938 vehicles/hour

Some hourly traffic counts were extremely high - over 760 vehicles per hour when the typical range is 143-357. Statistical tests identified 339,366 records as outliers (6.3% of all data).

In this case, we kept all outliers in the data as they are real data, not errors which represent legitimate rush hours, holiday travel, and special events. As removing them would hide important traffic patterns.

Problem 5: Duplicate Records

We looked for identical duplicate records that might have been entered twice by mistake. Checked all 5.4 million traffic records, 2,784 bus records, 2,786 passenger records, and 192 speed records. There were no duplicates, with 100% unique records across all datasets which indicated that the data was clean and required no action.

4. EXTERNAL DATA SOURCES

We are not using any external data sources at this time. The analysis relies exclusively on the four datasets provided by the Port Authority, totaling 5.4 million records. The Port Authority datasets are comprehensive enough for initial analysis of internal patterns and trends.

5. COMBINING/INTEGRATION THE DATASETS

The Challenge

We had four separate datasets with different time scales. So we used Python to tackle the issue.

Python Codes: [Data Exploration.pdf](#)

Our Solution: Aggregate Everything to Monthly

Step 1: Roll Up Traffic Data

The process began with a substantial dataset consisting of 5.4 million hourly traffic records. These records were grouped by year and month, allowing for a comprehensive understanding of traffic patterns over time. Monthly totals were calculated for key metrics, including TOTAL_TRAFFIC, TOTAL_VIOLATIONS, TOTAL_CASH, and TOTAL_EZPASS. As a result of this aggregation, the dataset was condensed to 149 rows, representing 12 years and 5 months of data.

Step 2: Roll Up Bus/Passenger Data

The focus shifted to bus and passenger data, which initially consisted of 2,784 weekly records for each dataset. To facilitate analysis, the weekly dates were converted to monthly periods, and all weekly data within each month was summed. The columns were then renamed to "PABT_Bus_Volume" and "PABT_Passenger_Volume" to reflect the aggregated information accurately. This step resulted in a more manageable dataset containing 53 rows, covering the period from December 2020 to May 2025.

Step 3: Prepare Speed Data

The speed data was already at a monthly level, eliminating the need for further aggregation. The month was extracted in a YYYY-MM format to ensure consistency with the other datasets. This step yielded a total of 192 rows, covering data up to March 2025. It is important to note that there were multiple rows per month, with each row corresponding to a specific facility-direction combination.

Step 4: Join Everything Together

Finally, the last step involved joining all the aggregated datasets together. This integration aimed to create a comprehensive dataset that encompasses traffic data, bus and passenger volumes, and speed metrics. By combining these datasets, a holistic view of transportation patterns and trends could be established, facilitating further analysis and decision-making.

We used the "left join" technique in this stage.

Step	Dataset Added	Result
Start	Traffic (149 months)	149 rows, 7 columns
Join #1	+ Speed data	~325 rows, 14 columns (12x rows for months with speed data)
Join #2	+ Bus data	~325 rows, 15 columns
Join #3	+ Passenger data	325 rows, 16 columns (final)

Due to the "one-to-many" join, the number of rows increased from 149 to 325. Each month of traffic data (one row) matched with multiple rows of speed data (12 facility-direction combinations). The traffic, bus, and passenger totals get repeated across all 12 rows for that month.

Example for January 2024:

- Traffic total: 9,273,213 (same on all 12 rows)
- Bus volume: 14,824 (same on all 12 rows)
- Speed at Lincoln Tunnel Eastbound: 21.0 mph
- Speed at Lincoln Tunnel Westbound: 26.8 mph
- Speed at GW Bridge Eastbound: 34.5 mph
- ...and so on for 6 facilities × 2 directions = 12 rows

Automation

We created a Python script to automate the processing of four datasets, which streamlines the workflow significantly. The script loads the datasets, cleans the data by fixing missing values and normalizing text, aggregates the data monthly, and joins everything into a single integrated dataset. It also exports this final dataset and generates summary statistics and validation reports, completing the entire process in about 30 seconds for 5.4 million records.

Python Codes: [Data Exploration.pdf](#)

The automation offers several key benefits. It ensures reproducibility by producing consistent outputs from the same inputs, maintains consistency in data cleaning rules across analyses, and improves efficiency by allowing quick reruns with new data. The well-documented code clarifies each step taken, and the automation is testable, enabling verification of results at each stage, which enhances the reliability of the analysis.

6. CREATING A SAMPLE FOR ANALYSIS

For this project, we took two different approaches depending on the analysis goal.

Approach #1: Sampling the Raw Hourly Data

When working with the original 5.4 million hourly records (for detailed visualizations like the traffic charts shown earlier), we created a sample to speed up processing. Our goal here was to create a smaller representative sample for quick exploration and charting so we used the stratified random sampling approach.

The data is grouped by facility, lane, and date. From each group, 3 records are randomly picked. This is repeated for every combination, and all the picked records are

combined into one sample. A random seed of 42 is used so the same sample is produced every time the code is run.

This approach ensures that every facility, lane, and date gets represented in the sample, not just the busy or recent ones. Simple random sampling could easily miss smaller facilities entirely, so this grouped method gives a more balanced and diverse picture of the data.

The original dataset had over 5 million hourly records. After sampling, the result is around 50,000 to 100,000 records, which is about 98% smaller. This makes the data much faster to work with for quick charts and early pattern detection, reducing processing from minutes to seconds.

Approach #2: Monthly Aggregation (Our Final Dataset)

Unlike the sampled datasets, the final integrated dataset used all 5.4 million records without leaving anything out. Instead of randomly picking records, all hourly records within each month were added up to create one monthly total (aggregation). Sampling picks a subset of records and discards the rest. Aggregation keeps 100% of the information but compresses it into summaries. So no data was lost here, it was simply condensed from hourly records into monthly totals.

The final dataset has 325 rows in total, which is significantly small in size but contains the full picture of everything in the original 5.4 million records.

7. FINAL DATASET FOR ANALYSIS

Specification	Details
Number of rows	325 rows (149 unique months, but multiple rows per month when speed data exists)
Number of columns	16 variables
Time coverage	January 2013 through May 2025 (12 years, 5 months)
Original records processed	5,383,378 traffic + 2,784 bus + 2,786 passengers + 192 speed

Strengths of This Dataset

This dataset boasts several significant strengths that enhance its utility for analysis. First, it provides a long time series consisting of 149 consecutive months, covering over 12 years, which enables thorough trend analysis and the detection of seasonal patterns. The quality of the data has been validated, with all 5.4 million records checked to ensure that totals match breakdowns 100%. Additionally, the dataset offers multiple perspectives for analysis, allowing users to explore traffic patterns, violations, payment methods, and vehicle types. The data is clean and consistent, as all text has been standardized and missing values have been handled uniformly. Furthermore, the dataset is ready to use in standard CSV format, making it compatible with various tools like Excel, Python, R, Power BI and Tableau.

Known Limitations

Despite its strengths, the dataset also has notable limitations. One significant issue is the gap in PABT data, which only covers 4.5 years of bus and passenger data compared to the 12+ years of traffic data available. Additionally, the speed data is scarce, with only three months available (March to May 2025), limiting the ability to analyze congestion trends over time. The presence of row duplication in the speed data creates 12 rows per month for each facility-direction combination, complicating some analyses. Moreover, the dataset combines data from multiple facilities, preventing comparisons of individual bridges or tunnels for traffic.